Manual

HYDRA Workflow-Designer
MDS-HWD 8.1

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Contents

1 HYDRA Workflow-Designer

With HYDRA Workflow designer you are able to create new workflow and modify existing standard workflows or customized individual workflows.

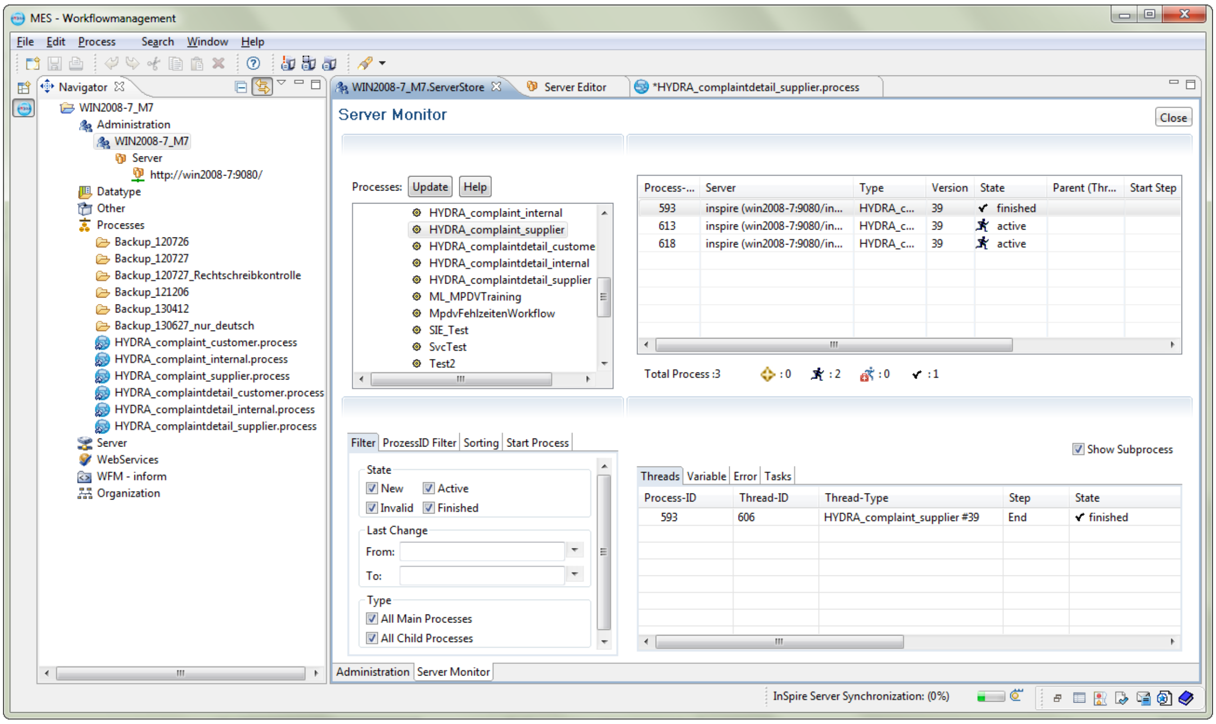
This document initially supplies background information of the HYDRA Workflow-Designer and the definition how to define a workflow.

2 Workflow Designer

Overview

MES workflow management enables the mapping of processes by means of a workflow. The workflows used for this are instantiated and thus started by HYDRA. In order for a workflow to be used, it must be created in a previous step in the workflow designer and assigned with appropriate parameters.

Workflow designer screenshot:



The workflow designer is not integrated in MOC and is implemented on the basis of Eclipse. In addition to the administration of the workflow management server, it also provides the interface for creating and deploying workflows.

In the initially installed standard status, the components required for using the workflow management and the workflow management database are found on the HYDRA server. This can be opened here via a link in the HYDRA administration file and/or via the Start menu by using the MES workflow management - designer link.

Prerequisite

JAVA 6 must be installed on the PC where the MES workflow designer is to be used.

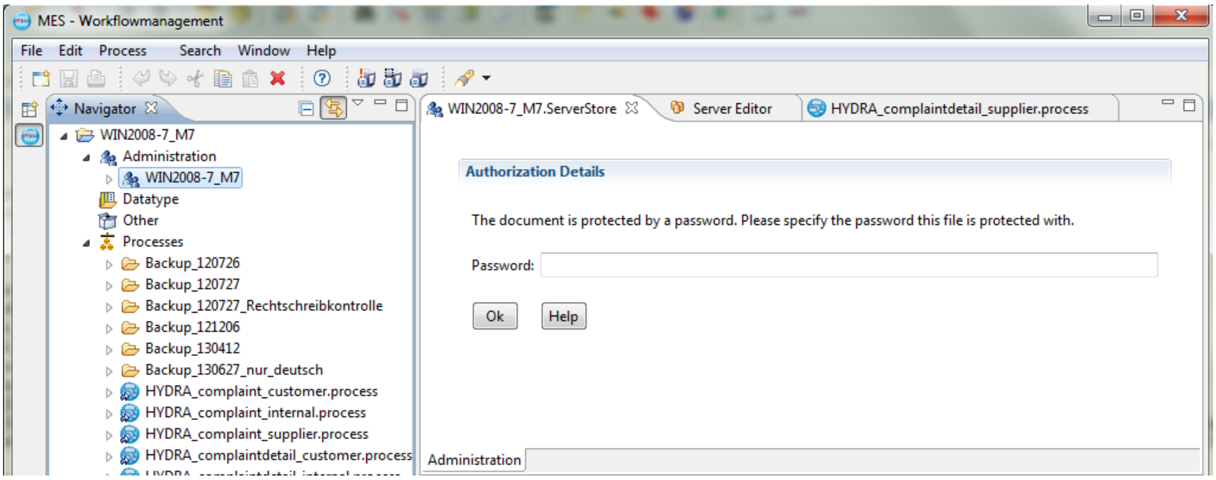
The server-side installation of workflow management must have been performed by MPDV commissioning. The server address and the workflow management port must be known. The password of the workflow administration user (e.g. admin) must be known.

The user must be familiar with the general characteristics of the workflow designer:

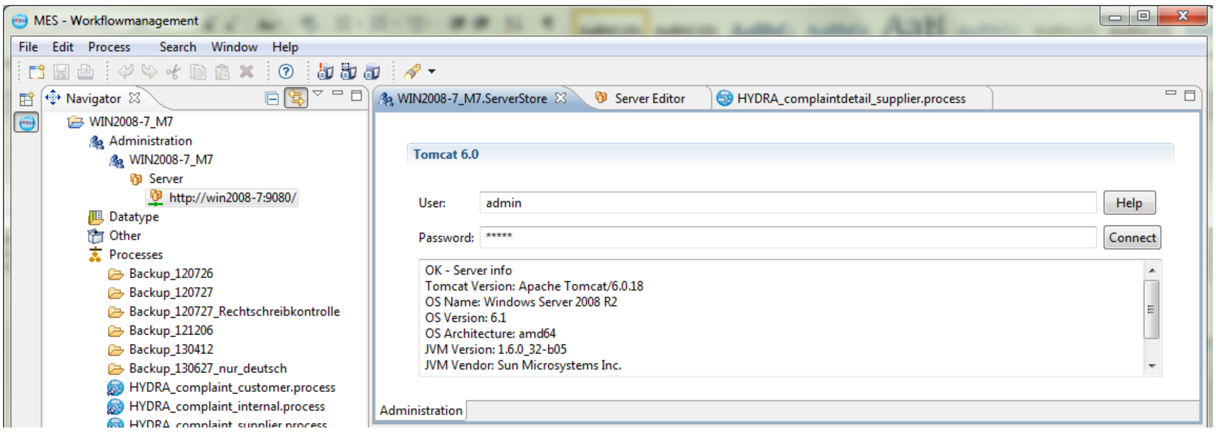
- Process server monitoring
- Process design
- Process deployment

2.1 Starting the Workflow Designer

When opening the workflow designer and double-clicking the Tomcat menu item (or the assigned name), the password assigned for the administration document must be entered and subsequently be confirmed by pressing the OK button so that a link to the workflow management server can be established.



Upon successful entry and connection to the workflow management server, the following information is displayed:



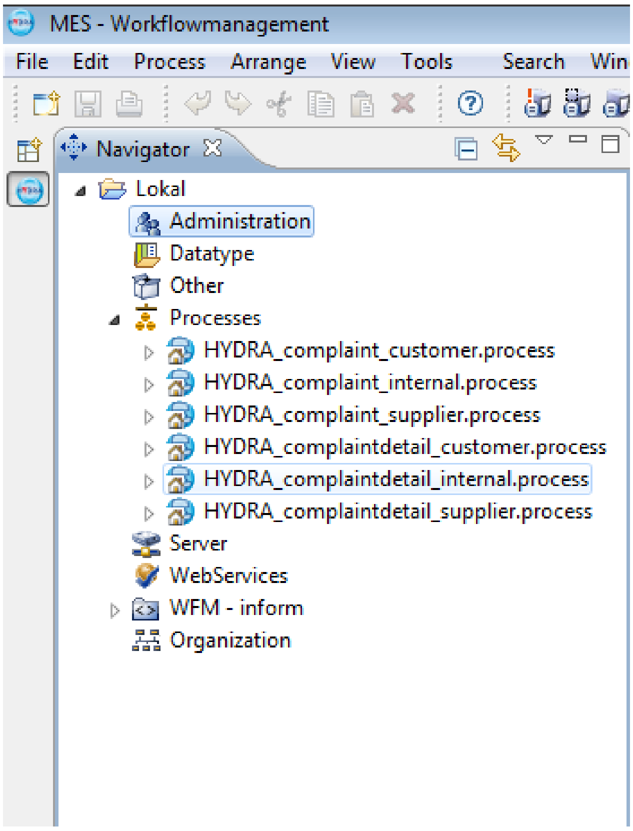
Once the connection has been made successfully, the following sub-items are found under the server entry (in this example under the entry: http://win2008-7:9080):

- WFM – Admin Server
  The sub-items in the WFM – Admin Server include the master data and/or settings required for the process of the workflows. The master data such as the user, however, are retrieved from HYDRA and synchronized with the workflow management system.
- WFM – Processes Server: inspire
  The processes server includes all deployed workflows and the associated required parameters.

2.2 Workflows

Creation of a New Workflow

By double-clicking the Processes item (see screenshot) and/or a folder created in it, the entry screen for the name of the new process opens. The naming convention is to be observed (please refer to section 2.8).



After entering the name, the new process is shown in the list of processes and, once closed, can also be re-opened by double-clicking the process name.

Design Guidelines for HYDRA Workflows

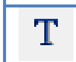
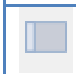
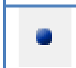
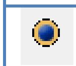
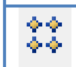
The following general conditions must be observed when designing a HYDRA workflow:

- The workflow must have precisely one starting point.
- The workflow must have at least one ending point. As an option, several ending points may be used.
- The workflow should, if possible, not call any sub-processes since these cannot be displayed in the workflow overview in MOC.

In the following sections, it is noted for several description fields that these serve 'technical documentation'.
The contents of these fields will not be visible for the editor of workflow tasks. For the designer of a workflow, however, these descriptions provide a quick overview as to what is hidden behind the variables, the e-mail templates and the work steps.
For this reason, this information should always be provided, since it significantly facilitates support and further development.

Workflow design

In addition to other options, the drawing surface for the workflow can be opened at any time by double-clicking on the workflow created. The following options for designing a workflow are available here:

Symbol	Designation
	Selection
	Shift view
	Zoom
	Text
	Positioning a pool
	Positioning the starting point of a process
	Positioning an ending point of a process
	Positioning a process step
	Creation of a sub-process call
	Creation of a multiple decision
	Positioning a decision element
	Positioning a 'Go to' / 'Merge' element

	Linking two elements

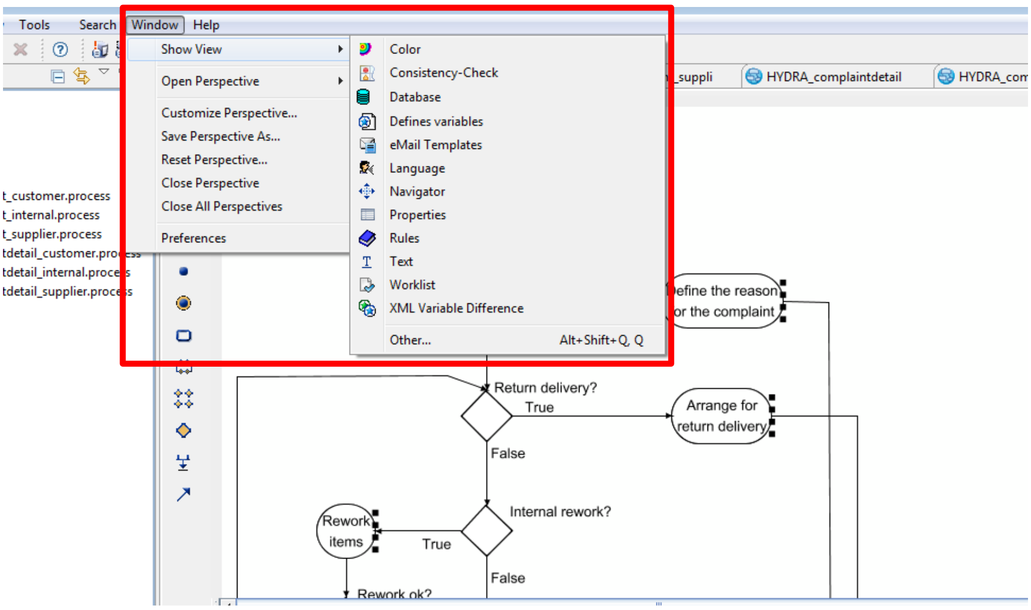
By drag & drop, the individual elements may be transferred to the drawing area for creating a workflow.

Properties for a Workflow

The following information can be maintained and functions called up for a workflow:

- Properties
  The properties of the currently highlighted workflow are displayed here
- Consistency check
  A current display of potential warnings, errors and information on the complete workflow is shown here
- Working list
  This includes the defined activities to be performed for the currently highlighted workflow step.
- Define variables
  In order for variables to be used in the individual actions of the workflow steps, they must first be defined and entered in the workflow. Please note: The naming conventions for variables are to be observed.
- E-mail template
  If an e-mail is to be sent using a defined e-mail template in an action, this template must first be defined on the workflow.
- Rules
  The defined rules relating to decisions in the workflow are maintained here. Only the rules defined in the current workflow step are displayed.

If one of the views specified is not available in the workflow designer, the required views can be displayed via the menu item Window → Show view (see screenshot).



Entering Variables

For using variables in the entire workflow, regardless of whether they are used in e-mail templates or for generic tasks, they have to be entered in the "Define variables" view on the workflow.

Here, variable names, the associated data type, a default value and a language-dependent description may be entered. For allocating variable names, the naming conventions are to be observed (see section 2.8).

In addition, a description may be entered. This is used for technical documentation and can be entered in several languages.

If a task screen in MOC is to be defined by means of the Java Inline Code, the fixed variable `taskData` must first be defined in the variables on a workflow ("Long string" data type).

Should parallel processes occur in a workflow, the related process step-specific variables `<Name_des_Prozessschritts>_taskData` must be defined.

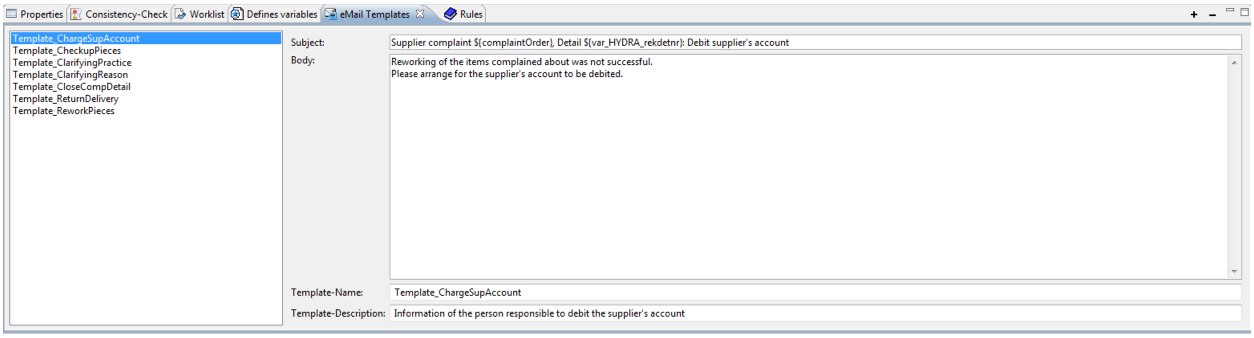
Entering E-Mail Templates

To be able to use e-mail templates in various actions, they must have been entered on the workflow in a first step. They can then be used and/or allocated in the individual actions via the template name.

When the e-mail is sent, the information in the header line is shown as the Subject and the information in the Mail Text cell is used as the e-mail text.

In addition to fixed texts, variables and line breaks can be used with the customary HTML formats for structuring, e.g.
 for a line break.

Variables can be used in the entire e-mail template by means of the syntax `${var_VariablenName}`.



When defining e-mail templates, the naming conventions are to be observed (see section 2.8).

The header and the e-mail text of the template can be entered in several languages.

In addition, a description may be entered. This is used for technical documentation and can be entered in several languages.

2.3 Elements of a Workflow Design

Starting Point

In the starting point properties, the Description parameter can be multilingual.

Ending Points

In the ending points properties, the Description parameter can be multilingual.

Process Steps

For some added process steps and/or decisions, the required parameters may be entered using the displayed views.
If a required view is missing, it can be displayed as described above.

In the process steps properties, the Description parameter can be multilingual.

Decision Elements

In the decision elements properties, the Description parameter can be multilingual.

Multiple Decisions

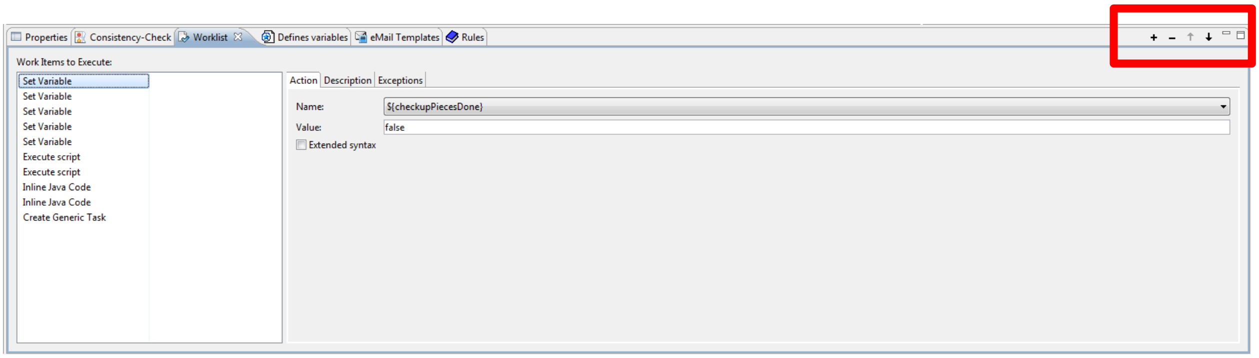
In the multiple decision properties, the Description parameter can be multilingual.

Links

In the links properties, the Description, Labeling 1 and Labeling 2 parameters can be multilingual.

2.4 Definition of the Working List of a Process Step

Using the working list context menu and/or the relevant buttons (see red marking), individual actions may be assigned to the process steps. Actions may be assigned several times or only once, depending on the action type.



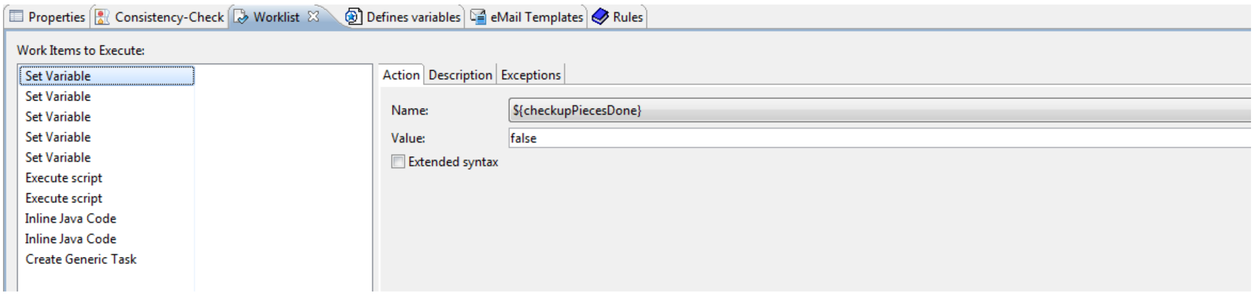
The following actions may be chosen:

- Set variable value
- Send e-mail
- Set xml variant
- Assign XML structure
- Inline Java Code
- Rule only
- Variable -> XML
- Create generic task
- Re-start workflow
- Start workflow
- Terminate workflow
- Call web service
- Execute simple EAI adaptor
- Execute script

For detailed information, please refer to the designer's Help function, which you can, for example, open directly via the Start menu under the Documentations item.

Set variable value

This action may be used to assign values to the variables entered in the workflow. The variable to be set may be explicitly set with a value entered in the Value field using the drop-down box. All variables which were entered on the current workflow in the "Define variables" view are available (see section 2.2 - Entering variables).

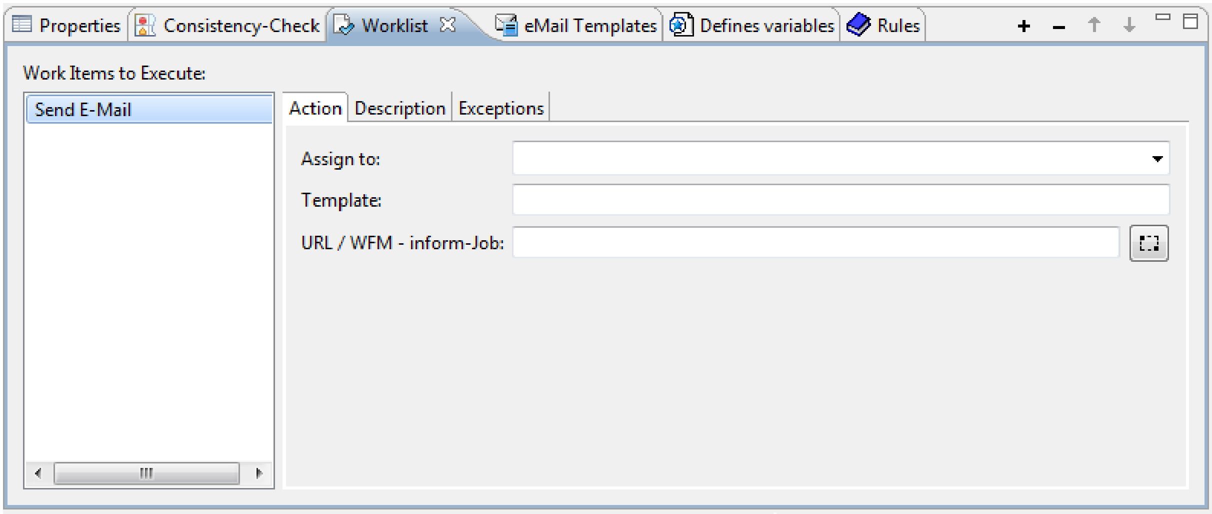


In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Send e-mail

Through the "Send e-mail" action, an e-mail can be sent to a user and/or a group or function by entering an e-mail template.

All e-mail templates available are defined in the "E-mail templates" view on the workflow and can be sent by assigning the name in a process step.



In the "Assign to" field, fixed values such as User abcd, but also variables may be used, which may include various addressees according to the setting of the variable in the process.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Please note: The technical prerequisites for sending e-mails are already met upon system startup.

Set xml variant

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

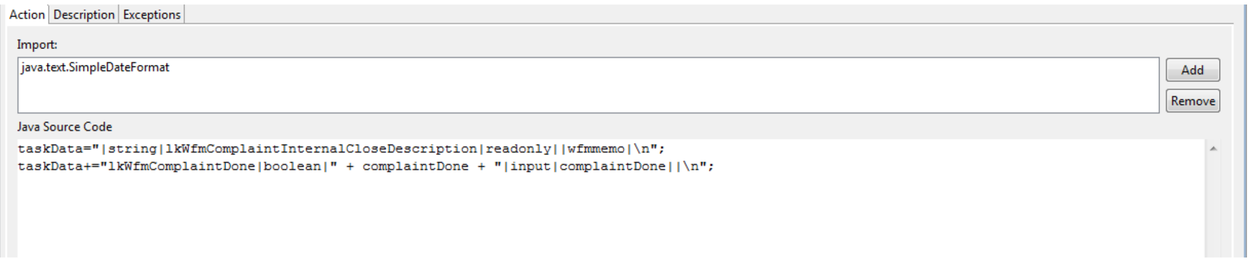
Assign XML structure

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Inline Java Code

The "Inline Java Code" action may be used to program functions by means of JAVA.



In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

By means of JAVA programming, for example, a description of the "Edit task" screen in MOC can be defined e.g. by supplying the variable `taskData`. This determines the appearance of the task screen here in the designer. The technical details of this solution are described below:

Prerequisite: The `taskData` variable must be added to the process with the "Long string" data type (see section 2.2 - Entering Variables).

Any field to be displayed in the task processing later is assigned in a separate line in this example for the `taskData` variable. The individual parameters which configure the field are to be separated by a pipe. The parameter set of a field is ended with `\n`.

Example for an Inline Java Code line:

```
taskData="Fertig?|boolean|" + taskDone + "|input|taskDone||\n";
```

The individual configuration parameters of a field defined by the `taskData` variable have the following meaning. The position here describes where the parameter is located in the pipe-separated list.

Pos.	Parameter	Description

1	Field label	This parameter determines the label in the MOC editing screen.
		
		If MOC task editing is to be available in one target language only, the label can be directly indicated here. Example:
		`taskData="Fertig?|boolean| " + done + "|input|done||\n"`
		
		If task editing, however, is to be used in several languages, an entry in the MOC dictionary can be referenced by a corresponding language key at this point. Example:
		`taskData="lkWfmDone|boolean| " + done + "|input|done||\n"`

2	Field data type	The field data type also determines the presentation in MOC. It must comply with the type of the associated variable (see parameter 5 and 3, where applicable). Example: `taskData="Palette|string|" + palId + "|input|palId||\n"` The following data types are available here: • **`boolean`** This data type represents a logic value which may assume the values `true` or `false`. In MOC, the field is displayed as a checkbox. • **`string`** This data type represents a character string. In MOC, the field is displayed as an input field. • **`datetime`** This data type represents a point in time (comprising date and time). In MOC, the field is displayed as an input field for times.

3	Value to be displayed	Value to be displayed in the field upon opening the editing dialog in MOC.
		
		At this point, a constant parameter may be entered. Example:
		
		`taskData="Palette|string|`**`P345`**`|input|palId||\n"`
		If this parameter is omitted completely, this can achieve the effect that the field is empty when opened.
		
		Alternatively, the contents of a defined variable may be displayed. Example:
		`taskData="Palette|string|`**`" + palId + "`**`|input|palId||\n"`

4	Editing/Display status	If the MOC user should be able to edit the field, `input` is used here. Example:
		`taskData="Palette|string||`**`input`**`|palId||\n"`
		
		Optionally, `readonly` is displayed. Example:
		`taskData="Palette|string||`**`readonly`**`|palId||\n"`

5	Input target	In the variable defined here, the contents of the associated field are saved when the task is saved in MOC. Example: `taskData="Palette|string|" + palId + "|input|`**`palId`**`||\n"`

6	Syntactic type	Syntactic type which defines how the value is to be displayed in MOC. Example:
		`taskData="Karte|string|" + kNr + "|input|kNr|`**`user_id`**`|\n"`
		
		By using the syntactic type `wfmmemo` you can achieve the effect that text information (`string` data type) is shown on several lines. Example:
		`taskData="|string|" + iTxt + "|readonly||`**`wfmmemo`**`|\n"`
		
		For entries of the syntactic type `wfmmemo`, an entry in the MOC dictionary can also be referenced via a corresponding language key instead of 'fixed contents'. This allows for optionally designing the contents of this multi-line field in several languages (e.g. in order to provide the user with notes on task handling). Example:
		
		`taskData="|string|lkWfmTaskDone|readonly||`**`wfmmemo`**`|\n"`

-	Line end	At the end of the parameters of a field, the line end `\n` is always required.

The first field to be indicated in the MOC editing screen of the task is assigned to the `taskData` variable as follows in this example:

`taskData=`*<Parameter of first field>*

If further fields are to be added, the relevant definitions may be added to the `taskData` variable as described below:

`taskData+=`*<Parameter of second field>*

`taskData+=`*<Parameter of third field>*

:

Using the `infoData` variable, a workflow detail dialog can be started, which will display fields for the entire workflow regardless of the current task. The field contents may also be edited in MOC as an option.

When configuring the fields of this dialog, the same functions as described above are applicable: However, for this dialog, the `infoData` variable instead of `taskData` is supplied with the field configurations.

When using **tasks in parallel**, the definition of fields for maintaining tasks by means of one variable would cause problems.
In order to avoid this, the definition of field configurations is not assigned to the `taskData` variable in this case. Instead, the name for this variable is formed as follows:

<Name_of_related_process step>`_taskData`

The name of the related process step can be taken from the properties of the process step.

Rule only

The "Rule only" action may be used to define an end condition for a process step behind one and/or several process steps.

The "Rule only" action can only be entered once at a process step. If an end condition has already been defined for a process step in another action, the "Rule only" action cannot be added any more.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Variable -> XML

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Create Generic Task

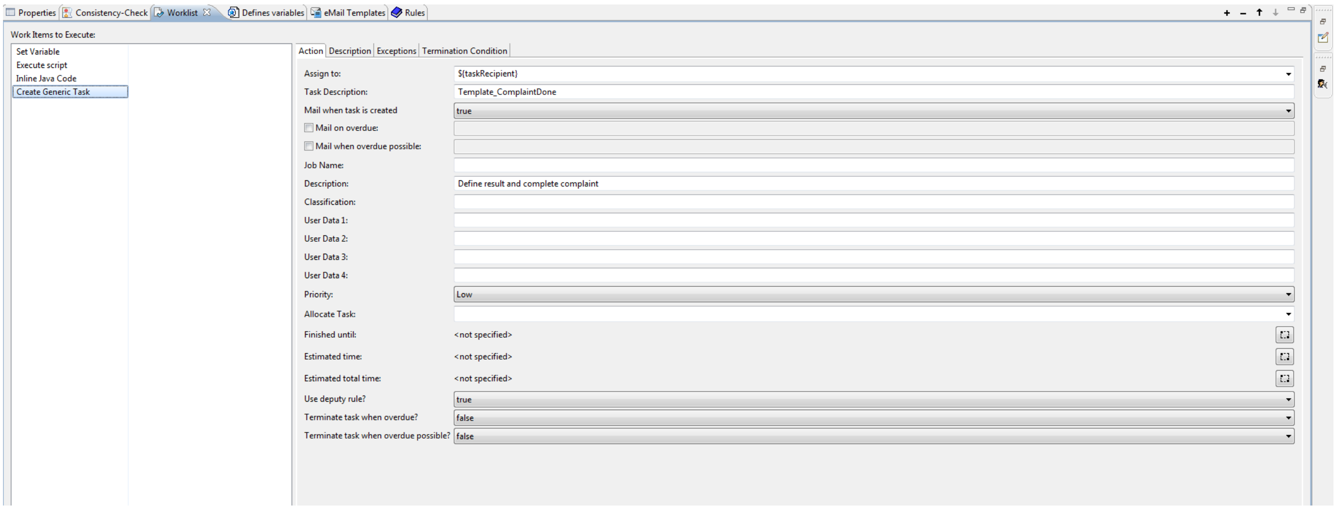
The "Create Generic Task" action can be used to enter tasks on the workflow steps, which must be performed before the workflow can be continued. Example: A responsible person shall decide whether or not a complaint will be processed.

For this purpose, the action for a generic task and the Java Inline Code action for the task screen in MOC must be entered.

In the Action tab, the following specifications may be defined for a generic task:

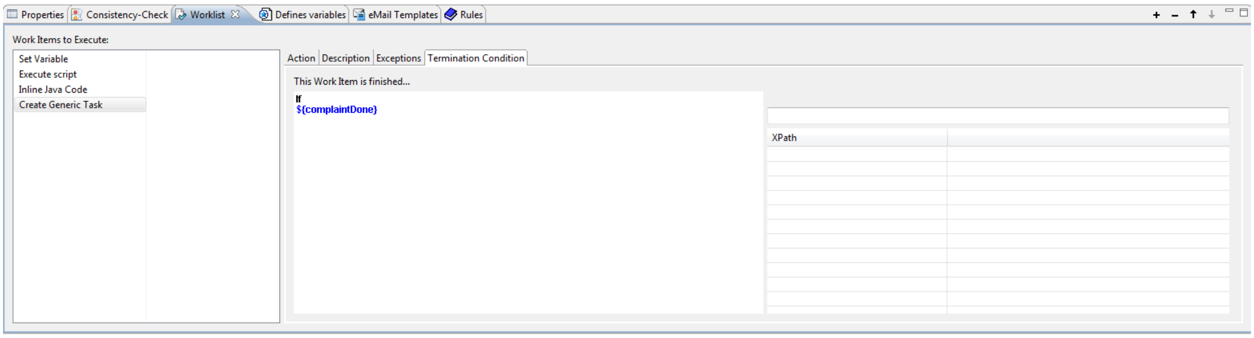
- Assign to:
  This field will assign the task, e.g. which user is assigned with the task. In addition to fixed values, variables may also be assigned, which are for instance determined in a previous action and/or in an upstream process step.
  Depending on whether a task is assigned to a USER, a GROUP or a FUNCTION, the type and the related name have to be defined e.g. by a variable.
- Task description
  A brief description of the task may be entered here in several languages. If an e-mail is to be sent (see next item), the task description is used as the name for the e-mail template.
- E-mail when creating the task
  Definition of whether an e-mail is to be sent when the task is created. If so, an e-mail template using the name of the task description is sought.
- E-mail if due date is exceeded
- E-mail if due date might be exceeded
- Job Name
- Description
  Multilingual field for task description
- Classification: <Future use>
  This field can be multilingual.
- User data 1-4: User-specific data may be entered in these fields.
- Priority
  The priority is shown in the task list on MOC and sorting/grouping is possible.
- Reserve task: <Future use>

- Done by
- Estimated duration
- Estimated total duration
- Apply absence management
  Shall defined absence management be applied?
- Terminate task if due date is exceeded?
- Terminate task if due date might be exceeded?



In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

A terminate condition for this process step may be defined under the "End condition" tab. In this example (see Screenshot), the End condition is reached if the `${var_ok}` variable is assigned with `true`. To prevent that this is the case from the start already, the value may explicitly be set to `false` by means of the "Set variable value".



Re-start workflow

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Start workflow

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Terminate workflow

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Call web service

Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

The description in the tab with the same name can be multilingual.

Execute simple EAI adaptor

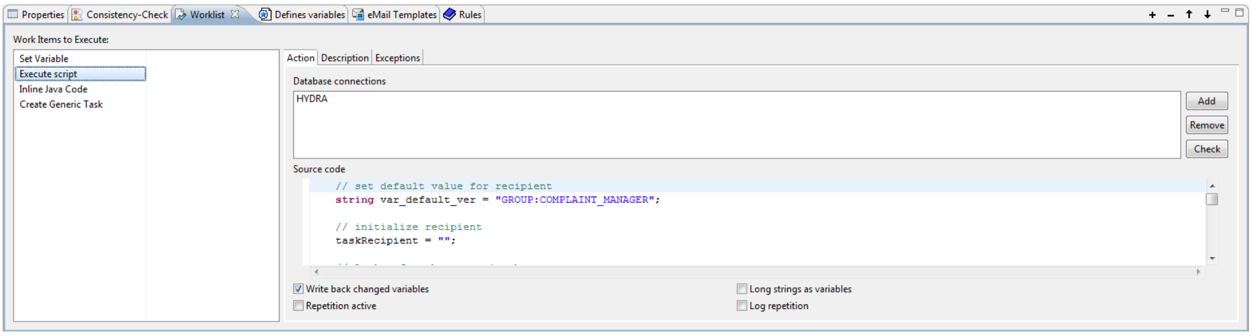
Currently not used for MPDV workflows. For information on this action type, please refer to the designer's Help function.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

Execute script

In the current process, HYDRA data saved in the database may be accessed from the workflow. For this purpose, the "Execute script" action type must be used. The HYDRA database is accessed via JDBC.

A database link must be added first. The Add button opens a list with the configured databases. In the following example, the database link "HYDRA" was selected.



In this example, other machine master data (designation and cost center) are read. The SQL statement is assigned to the `sel` string. At this point, the SQL keyword "select" is not indicated!

```
string sel = "bez_lang, kostenstelle from maschinen where masch_nr = '" + var_mnr + "'" ;
// Mit set_first kann nach einem erfolgreichen Select geprüft werden,
// ob Daten gefunden wurden.
if (select("mnr", sel, "HYDRA") && set_first("mnr"))
{
   // Schleife über die Zeilen.
   //do {
     var_bezl = get_value("mnr",0);
     var_kst = get_value("mnr",1);
   //} while (set_next("mnr"));
 }
 // Datenobjekt freigeben
 end_select("mnr");
 return true;
```

The do - while loop was omitted since only one data record is expected in the result data quantity. The machine designation and the cost center are assigned to the relevant process variables.

The "Apply modified variables" option must be selected.

In addition, a description may be entered in the same tab. This is used for technical documentation and can be entered in several languages.

2.5 Deploying Workflows

After completion of a workflow, it must be deployed via the workflow designer so that it is distributed on the server and can thus be executed. The relevant buttons are found in the toolbar:

- 

  Quick Deploy of modified ME workflow management resources – deployment is triggered for all modified workflows without any selection or confirmation
- 

  Quick Deploy of selected MES workflow management resources – after selecting the workflows to be deployed, deployment is triggered for the selected workflow.

- 

  Deployment of MES workflow management resources – after selecting the project and the workflows to be deployed, deployment is triggered for the selected workflows.

If the workflow to be deployed contains errors, deployment is not executed and the errors are listed.

Whether deployment is executed with existing warnings can be selected when deploying.

After deployment, the workflows are available on the server and in the workflow designer under the item WFM Process Designer: inspire → Process Definitions, and processes may be started and/or current processes may be viewed for the different versions.

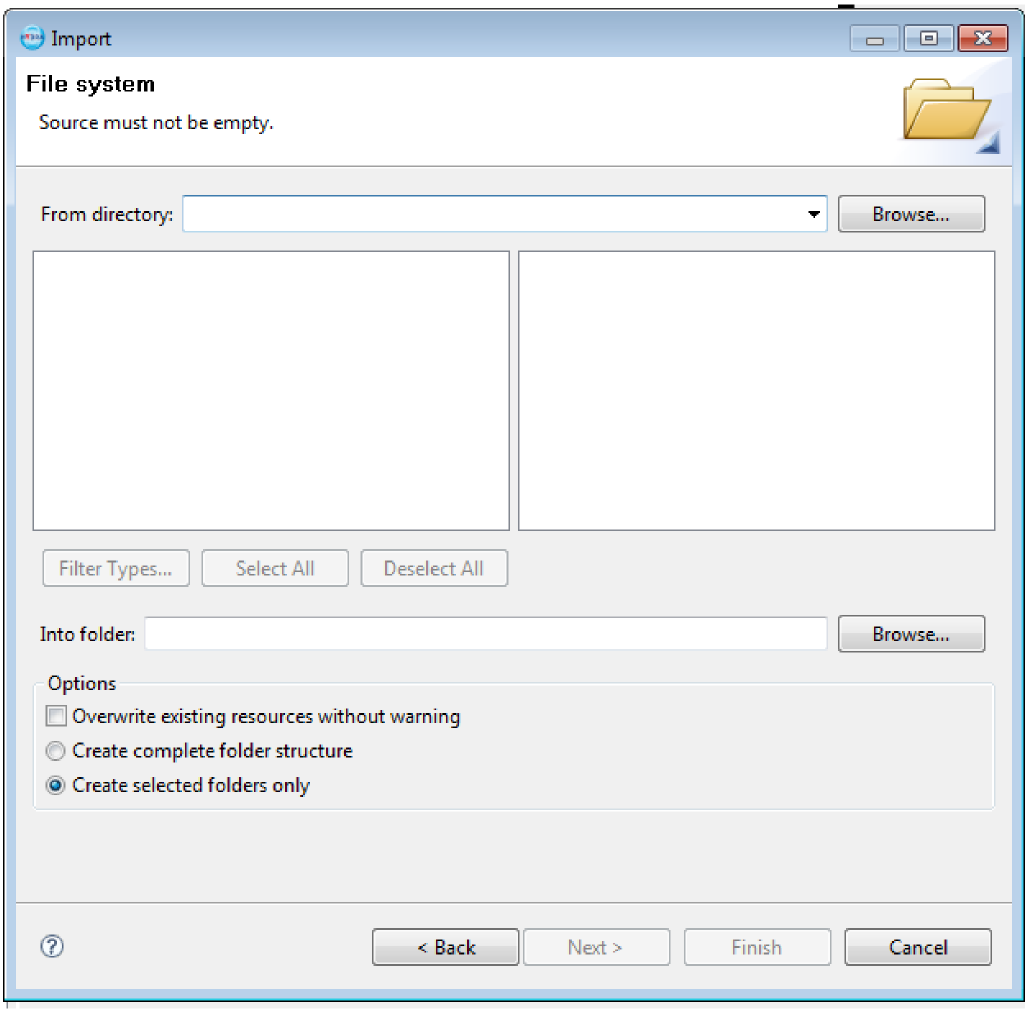
2.6 Import/Export of Workflows

Import of Workflows

In order for existing workflows to be edited in the workflow designer, these have to be imported first.

Under the File → Import to menu item, existing and exported workflows can be imported.

Screenshot:



The default workflows which can be used as default for system startup can be found under the following directory on the HYDRA server:

\\<HYDRA-Server>\<HYDRA-Verzeichnis>\products\wfm\

Export of Workflows

In order to save and/or distribute existing workflows, these may be exported from the workflow designer and filed on the target system by means of the File → Export to menu item.

2.7 Language / Multilingual input

The Language view can be used to enter designations in several languages in individual fields, e.g. a workflow step designation.

For this purpose, the field where the individual languages are required has to be maintained appropriately. If a language is not maintained, the default language will be displayed in the workflow designer.

Please note: The defined default language for the created workflows is German.

2.8 Naming Conventions

Naming Conventions for Workflows

The names of process steps and links must be English and follow the conventions described below:

- Start:
  `Start:`
- End:
  `End` or
  `End_`*<Description in PascalCase>*
  Example: `End_Success`
- Process steps:
  `Step_`*<Description in PascalCase>*
  Example: `Step_InvestigateTask`
- Decisions / Multiple decisions:
  `Decision_`*<Description in PascalCase>*
  Example: `Decision_Justified`
- Links:
  `Link_`*<Description in CamelCase>*
  Example: `Link_DoReworkCustomer`
  `Link_ReworkCustomerDone`
  `Link_ArrivalConfirmed`
- Splitting:
  `Split_`*<Description in CamelCase>*
  Example: `Split_UrgentCause`
- Merger:
  `Merge_`*<Description in CamelCase>*
  Example: `Merge_8dClose`

Character string variables whose contents will affect workflow control (e.g. simple multiple decision using precisely one variable) should have English contents, if possible.

The labels used for the associated elements in the taskData variable must be English and follow the conventions below:

- lkWfm_<Description in PascalCase>
  Example: lkWfmComplaintDataCollected

If the name of the variable was selected appropriately, there is nothing wrong with setting the variable name behind the prefix lkWfm. If a generic task was assigned to a process step, the description of this task can usually correspond to the text of the process step in the sequence chart. The description of a generic task usually matches the description of the associated process step.

Naming conventions for e-mail templates

The header of the e-mail template of a generic task usually includes the description of the generic task. As a prefix, the respective context should also be included. The label for entering a variable (in the context of a task) in HYDRA should usually match the description of the relevant task.

The names of the e-mail templates must be English and follow the conventions described below:

- Template_<Description in PascalCase>
  Example: Template_CollectData

Naming Conventions for Variables

The names of the variables must be English and follow the conventions described below:

- General variables
  <Description in CamelCase> Example: `complaintDataCollected`
- Variables whose contents are provided by HYDRA upon instantiation of a workflow
  `var_HYDRA_`<Name of field in HYDRA in lower case> or
  `var_HYDRA_`<Name of parameter in HYDRA in PascalCase>
  Example.: `var_HYDRA_reknr`
  `var_HYDRA_InspectionStep`

2.9 Documentation of Created Workflows

For the documentation of edited and/or created workflows, 2 documentation types are distinguished:

- User-related documentation on workflow
- Technical workflow documentation

User-related Documentation on Workflow

User-related documentation should primarily contain the defined workflows with individual workflow steps from the end user's perspective. It must be ensured that a graphic representation of the entire process on the one hand, but also a definition of the individual process steps with the underlying actions and/or tasks on the other hand, are provided for the end user.

Technical Workflow Documentation

Via the Process → Generate document menu item, technical documentation can and/or must be prepared for the created workflows. The contents to be used for documentation may be selected in individual steps. The following contents are available here:

- Variables

- E-mail templates
- Merger
- Decision
- Sub-process
- Selection
- Step
  - Step
  - Working list
  - Action
  - Exception

Documentation should be prepared as rtf, since in addition to the text components a screenshot of the designed workflow will subsequently have to be integrated in the documentation